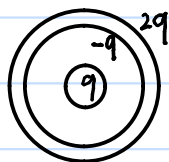


第十二章作业

教材作业

$$3. \varphi_3 = \frac{q}{4\pi\epsilon_0 R_3}, \varphi_2 = \frac{-q}{4\pi\epsilon_0 R_2}, \varphi_1 = \frac{2q}{4\pi\epsilon_0 R_1}$$

$$\varphi_0 = \varphi_1 + \varphi_2 + \varphi_3 = \frac{q}{4\pi\epsilon_0} \left(\frac{2}{R_1} + \frac{1}{R_3} - \frac{1}{R_2} \right)$$



7. 设圆柱电荷线密度为 λ

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$V_2 - V_1 = \int_{R_1}^{R_2} \frac{\lambda}{2\pi\epsilon_0 r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}$$

$$\text{则 } \lambda = 2\pi\epsilon_0 (V_2 - V_1) / \ln \frac{R_2}{R_1}$$

$$\text{故 } V(r) = V_1 + \int_{R_1}^r \frac{\lambda}{2\pi\epsilon_0 r} dr = V_1 + (V_2 - V_1) \frac{\ln \frac{r}{R_1}}{\ln \frac{R_2}{R_1}} \quad (R_1 < r < R_2)$$

$$8. dR = \frac{1}{y} \cdot \frac{dl}{\pi r^2}$$

$$R = \int_0^h \frac{1}{y} \cdot \frac{dl}{\pi \left(r_1 + \frac{r_2 - r_1}{h} l \right)^2} = \frac{1}{\pi y} \cdot \frac{h}{r_2 - r_1} \left(-\frac{1}{r_1 + \frac{r_2 - r_1}{h} l} \right) \Big|_0^h = \frac{1}{\pi y} \cdot \frac{h}{r_2 - r_1} \cdot \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{h}{\pi y r_1 r_2}$$

$$12. (1) \mathcal{E}_1 = i(R_1 + R_2 + R_3 + R_{i1} + R_{i2}) + \mathcal{E}_2$$

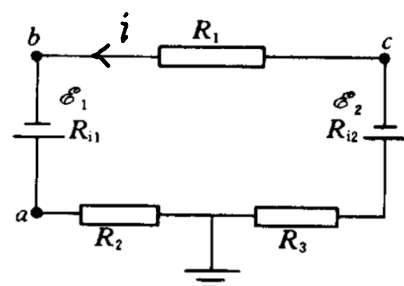
$$i = \frac{18}{9} = 2A$$

$$(2) \varphi_a = iR_2 = 2V$$

$$\varphi_b = i(R_2 + R_{i1}) - \mathcal{E}_1 = -18V$$

$$\varphi_c = -i(R_3 + R_{i2}) - \mathcal{E}_2 = -14V$$

$$(3) U_{\mathcal{E}_1} = \varphi_a - \varphi_b = 20V$$

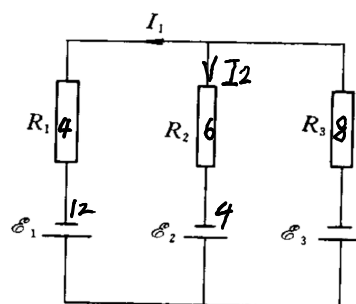


$$13. I_1 R_1 - \mathcal{E}_1 = I_2 R_2 - \mathcal{E}_2$$

$$I_2 = -1A$$

$$I_1 R_1 - \mathcal{E}_1 = -(I_1 + I_2) R_3 - \mathcal{E}_3$$

$$\mathcal{E}_3 = 14V$$



系列化问题

2. (1) 外球本身不带电, $q_{\text{外}} = 0$

$$E \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\varphi_{\text{外}} = \int_{r_1}^{+\infty} E dr = \frac{q}{4\pi\epsilon_0 r_1}$$

(2) 由高斯定理, $q_{\text{外}} = -q$

接地, $\varphi_{\text{外}} = 0$

$$(3) E = \begin{cases} \frac{q_{\text{内}}}{4\pi\epsilon_0 r^2}, & R_2 < r \leq R_1 \\ \frac{q_{\text{内}} - q}{4\pi\epsilon_0 r^2}, & r > R_1 \end{cases}$$

$$\varphi_{\text{内}} = \int_{r_2}^{+\infty} E dr = \frac{1}{4\pi\epsilon_0} \left(\frac{q_{\text{内}}}{r_2} - \frac{q}{r_1} \right) = 0$$

$$q_{\text{内}} = q \frac{r_2}{r_1}$$

$$\varphi_{\text{外}} = \int_{r_1}^{+\infty} E \cdot dr = \frac{-q}{4\pi\epsilon_0 r_1} \left(\frac{r_2}{r_1} - 1 \right)$$

$$\Delta q_{\text{内}} = q \left(\frac{r_2}{r_1} - 1 \right), \quad \Delta \varphi_{\text{外}} = \frac{r_1 - r_2}{4\pi\epsilon_0 r_1^2} q$$

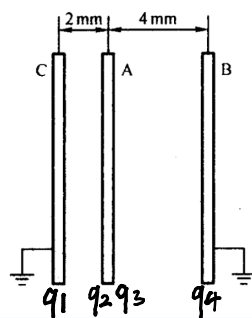
$$4. \begin{cases} q_2 + q_3 = 3 \times 10^{-7} \text{C} \\ q_1 + q_2 = 0 \\ q_3 + q_4 = 0 \\ \frac{q_1}{q_4} = \frac{C_{AC}}{C_{AB}} = 2 \end{cases}$$

$$(1) q_3 = 10^{-7} \text{C}, q_2 = 2 \times 10^{-7} \text{C}$$

$$q_B = q_4 = -10^{-7} \text{C}, q_C = q_1 = -2 \times 10^{-7} \text{C}$$

$$(2) C_{AC} = \frac{\epsilon_r S}{4\pi k d_{AC}}$$

$$\varphi_A = U_{AO} = \frac{q_C}{C_{AC}} = \frac{-q_C d_{AC}}{\epsilon_0 \epsilon_r S} \approx 2.26 \times 10^3 \text{V}$$



8. (1) $-q_A - q_C = Q$

$$E = \begin{cases} \frac{q_A}{2\pi\epsilon_0 r L}, & R \leq r < 2R \\ \frac{-q_C}{2\pi\epsilon_0 r L}, & 2R < r \leq 4R \end{cases}$$

$$U_{AC} = \int_R^{4R} E \cdot dr = \frac{\ln 2}{2\pi\epsilon_0 L} (q_A - q_C) = 0$$

$$q_A = q_C = -\frac{Q}{2}$$

$$(2) \varphi_B = -\int_R^{4R} E \cdot dr = \frac{Q \ln 2}{4\pi\epsilon_0 L}$$

$$(3) U_{AB} = U_{CB}, q_A = q_C$$

$$C_{AB} = C_{CB} = \frac{q}{U} = \frac{2\pi\epsilon_0 L}{\ln 2}$$

$$C = C_{AB} + C_{CB} = \frac{4\pi\epsilon_0 L}{\ln 2}$$

9. 由高斯定理, 内表面带电 $-q$, 外表面带电 $Q+q$

$$(1) \varphi_0 = \frac{q}{4\pi\epsilon_0 r} - \frac{q}{4\pi\epsilon_0 R_1} + \frac{Q+q}{4\pi\epsilon_0 R_2}$$

$$(2) E \cdot 4\pi R^2 = \frac{Q+q}{\epsilon_0}$$

$$E = \frac{Q+q}{4\pi\epsilon_0 R^2}$$

10. (1) $\frac{\sigma_1 R_1}{\epsilon_0} = \frac{\sigma_2 R_2}{\epsilon_0}$

$$\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

$$(2) U = \frac{\sigma_1 R_1}{\epsilon_0}$$

$$C = \frac{Q}{U} = \frac{\sigma_1 4\pi R_1^2 + \sigma_2 4\pi R_2^2}{\frac{\sigma_1 R_1}{\epsilon_0}} = 4\pi\epsilon_0 (R_1 + R_2)$$

13. (1) $j = \frac{I}{S} = \gamma_1 E_1 = \gamma_2 E_2$

$$E_1 = \frac{I}{\gamma_1 S}, \quad E_2 = \frac{I}{\gamma_2 S}$$

$$(2) U_{AB} = E_1 d_1 = \frac{Id_1}{\gamma_1 S}$$

$$U_{BC} = E_2 d_2 = \frac{Id_2}{\gamma_2 S}$$

$$(3) j = \gamma E, \quad \gamma \gg \gamma_1, \gamma_2$$

$$E \ll E_1, E_2$$

作过A的柱形高斯面

$$E_1 \cdot S = \frac{\sigma_A S}{\epsilon_0}, \quad \sigma_A = \frac{\epsilon_0 I}{\gamma_1 S}$$

$$E_2 \cdot S = \frac{\sigma_C S}{\epsilon_0}, \quad \sigma_C = \frac{\epsilon_0 I}{\gamma_2 S}$$

$$(E_2 - E_1) \cdot S = \frac{\sigma_B S}{\epsilon_0}, \quad \sigma_B = \frac{\epsilon_0 I}{S} \left(\frac{1}{\gamma_2} - \frac{1}{\gamma_1} \right)$$

补充作业

1. 在表面取微元 ΔS 并作高斯面

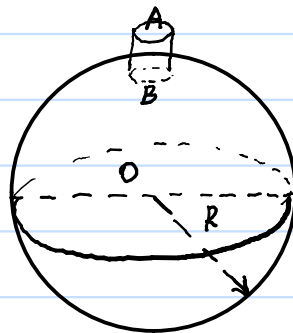
设 ΔS 在此处产生场强 E_1 , 其余部分产生场强 E_2 .

$$\begin{cases} E_A = E_1 + E_2 = \frac{Q}{4\pi\epsilon_0 R^2} \\ E_B = E_2 - E_1 = 0 \end{cases} \Rightarrow \begin{cases} E_1 = \frac{Q}{8\pi\epsilon_0 R^2} \\ E_2 = \frac{Q}{8\pi\epsilon_0 R^2} \end{cases}$$

$$\text{单位面积受力密度 } \vec{f} = \frac{\sigma \Delta S \cdot E_2}{\Delta S} \vec{e}_r = \frac{Q^2}{32\pi^2 \epsilon_0 R^4} \vec{e}_r$$

$$\text{下半球对上半球电力 } F = \iint_{S_1} \vec{f} \cdot d\vec{S} = \iint_S f \cdot dS \cdot \cos\theta = f \cdot \pi R^2 = \frac{Q^2}{32\pi\epsilon_0 R^2}$$

$$\text{故至少需要 } F = \frac{1}{32\pi\epsilon_0} \cdot \frac{Q^2}{R^2}.$$



$$2. (1) \begin{cases} \varphi_0 = \frac{q}{4\pi\epsilon_0 d} + \frac{q_A}{4\pi\epsilon_0 R_A} + \frac{q_B}{4\pi\epsilon_0 R_B} = 0 \\ q_B = 0 \end{cases}$$

$$q_A = -q \cdot \frac{R_A}{d}, \quad q_B = 0$$

$$(2) \quad q_{\text{内}} = -q_A = q \frac{R_A}{d}$$

$$q_{\text{内}} \text{ 与 } q_A \text{ 对 } O \text{ 的电势 } \varphi'_0 = -\frac{q}{4\pi\epsilon_0 d} + \frac{q}{4\pi\epsilon_0 d} \cdot \frac{R_A}{R_B} = \frac{q}{4\pi\epsilon_0 d} \left(\frac{R_A}{R_B} - 1 \right)$$

$$\text{而 } \varphi_0 = 0, \text{ 有 } \varphi_B = -\varphi'_0 = \frac{q}{4\pi\epsilon_0 d} \left(1 - \frac{R_A}{R_B} \right)$$

附加题

1. 电荷分布存在如下约束:

$$\begin{cases} \iint_S \sigma \cdot dS = Q \\ \varphi(r) = \text{const.} \end{cases}$$

考虑半径为 R , 带电 Q 的球壳

在球壳内有 $E(\vec{r}) = 0, \varphi(\vec{r}) = \text{const.}$

将球壳电荷投影到大圆上, 仍有 $\iint_S \sigma dS = Q$

又 $\Delta E_1 = \Delta E_2$, 有 $\Delta E \cos \theta = \Delta E_2 \cos \theta$

则平行圆面方向 $E_r = 0$, 也有 $\varphi(r) = \text{const.}$

圆盘的电荷分布与球面投影完全一致.

考虑关于圆面对称的两面元

$$\begin{cases} 2\sigma_0 \Delta S = \sigma \Delta S \sin \theta \\ \sin \theta = \frac{\sqrt{R^2 - r^2}}{R} \end{cases}$$

$$\text{故 } \sigma(r) = 2\sigma_0 \frac{R}{\sqrt{R^2 - r^2}} = \frac{Q}{2\pi R \sqrt{R^2 - r^2}}$$

参考文献: 熊建平, 导体薄圆盘的电荷分布, 大学物理, 1999.5

